



Divit Mittal

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Kaggle: <https://www.kaggle.com/divitmittal>

HuggingFace: <https://huggingface.co/divitmittal>

Website: <http://www.divit.qezta.com/>

SKILLS

Machine Learning & Data Science (Mathematics)

Deep Learning: Keras & PyTorch / Machine Learning: scikit-learn, tensorflow / LLM: Transformers, RAG, Encoders, & Decoders / Big Data: Hadoop, Spark, Hive, Kafka / Data Dashboarding: MS PowerBI / SysAdmin: Linux & macOS (systemd, launchd, kernel configuration) / Image(DL): Vision Transformers, Semantic Segmentation, GANs, Siamese / Computer Vision: OpenCV, ITK

DevOps & Linux SysAdmin

NixOS Expert / IaC: nix, ansible & terraform / Shells: bash, zsh, fish, pwsh / CI/CD: Github Actions & Hydra / Cloud: AWS (Lambda, S3, Glue, Sagemaker, API Gateway) / Declarative Agent AI Workflows / VCS: Git & Jujutsu / Virtualization & Containerization: Proxmox, KVM, Docker & Podman

WORK EXPERIENCE

Manipal University – Jaipur, Rajasthan, India

Name of unit or department: Department of Applied Statistics and Data Science - **Business or sector:** Education

Links <https://huggingface.co/spaces/divitmittal/HybridTransformer-MFIF> | <https://kaggle.com/datasets/divitmittal/lytro-multi-focal-images>

Funded Deep-Learning Research Position

[15/05/2025 – 01/09/2025]

Research on deep learning architecture combining **FocalViT** and **CrossViT** transformers for **multi-focus image fusion**, addressing the challenge of merging images with different focal planes into a single all-in-focus result. Achieved commendable fusion quality on benchmark datasets, demonstrating improved preservation of high-frequency details compared to CNN-based approaches.

Remote

Open Source & Community Contributions

[01/07/2024 – 01/10/2024]

Active contributor to the functional programming ecosystem, particularly within the Nix community. Participated in Nix consulting with **Tweag**, **ModusCreate**, and **Blockfrost**, providing expertise on complex NixOS configurations, package derivations, and infrastructure automation for enterprise clients.

UbiOps (Yes!Delft) – Delft, Netherlands

DevOps & System Administrator

[15/12/2021 – 28/02/2022]

Administered containerized environments using Docker/Kubernetes for multi-tenant ML workloads, implemented CI/CD pipelines for automated model deployment, and maintained cloud infrastructure across AWS/GCP.

EDUCATION AND TRAINING

Bachelor of Technology in Data Science & Engineering

Manipal University [10/09/2022 – Current]

City: Jaipur, Rajasthan | Country: India | Website: <https://jaipur.manipal.edu/> | Final grade: ~8/10 | NQF Level: NCC5

Bachelor of Science in Computer Science & Engineering

Delft University of Technology [01/09/2021 – 01/09/2022]

City: Delft | Country: Netherlands | Website: tudelft.nl | Final grade: ~8/10 | Type of credits: ECTS | Number of credits: 20

Bachelor of Computing in Computer Science

National University of Singapore [01/08/2020 – 28/08/2020]

City: Singapore | Country: Singapore | Website: <https://www.nus.edu.sg/>

All India Senior School Certificate Examination

Central Board of Secondary Education (CBSE) [31/03/2018 – 12/07/2020]

Address: Delhi Public School, R.K. Puram, New Delhi (India) | Final grade: A1 : ~96% | NQF Level: NSQF level 4 - (NCC 2)

All India Secondary School Examination

Central Board of Secondary Education (CBSE) [31/03/2016 – 28/05/2018]

Address: Bhatnagar International School, Bhera Enclave Paschim Vihar, New Delhi (India) | Final grade: A1 : ~97% | NQF Level: NSQF level 2 - (NCWP 2)

LANGUAGE SKILLS

Mother tongue(s): Hindi

Other language(s):

German/Deutsch (Goethe-Zertifikat B2)

LISTENING B2 READING B2 WRITING B2

SPOKEN PRODUCTION B2 SPOKEN INTERACTION B2

English (IELTS)

LISTENING C2 READING C2 WRITING C1

SPOKEN PRODUCTION C1 SPOKEN INTERACTION C2

PROJECTS

[04/2024 – Current]

OS-nixCfg: State-of-the-art Infrastructure as Code(IaC) configuration Nix, nix-darwin, **NixOS**, home-manager, etc.

configurations to attain my **deterministically reproducible** OS across heterogeneous platforms, written in the **functional** "Nix" expression language. Maintains:

- Unified configuration for **4+** platforms (macOS via nix-darwin, Android via nix-on-droid, NixOS, WSL2) achieving complete reproducibility.
- Starred by DRW employees, demonstrating **industry validation** of advanced Nix expertise.
- Implements flake-parts modular architecture, agenix/ragenix secrets management, nix-topology network visualization, and CI/CD with GitHub Actions.
- Custom package derivations, overlays, & home-manager modules spanning development tools, GUI applications, and system services contributions to the **ecosystem**.

Link: <https://github.com/DivitMittal/OS-nixCfg>

[02/2024]

Programming Languages Documentation Conversational AI @ IEEE MUJ RAG-enhanced LLaMa-like transformer, written from scratch for assistance on programming languages/frameworks' documentations via a LLM with a **streamlit** front-end.

- Pre-trained **LLaMa-like transformer** with capability from **tokenization** to **embedding**(positional & word2vector) to output probabilities.
- **RAG** was implemented via **Qdrant** vector database on a **DigitalOcean Docker** container.

Link: <https://github.com/DivitMittal/DocAssist-LLM>

[04/2025]

Blinkit Customer Churn Analysis

- Exploratory Analysis: Employed **Sweetviz** for comprehensive automated data profiling and feature correlation analysis
- Impact: Identified actionable insights for customer retention campaigns, enabling data-driven business decisions
- Predictive Modelling: Developed churn prediction models with interpretable feature importance for business strategy
- Tech Stack: Python, H2O.ai AutoML, PowerBI, Sweetviz

Link: <https://github.com/DivitMittal/Blinkit-Churn-Analysis>

[10/2023]

Generation of "Hazard Map" for safely navigating a lunar lander Is implemented by utilising the sensory data from the **ISRO's** Chandrayaan-2 TMC2 & OHRC payloads for **SIH**(Smart India Hackathon). The project entails wrangling multiple data in form of multispectral high-res. imagery & stereo-triplet panchromatic data of lunar terrain. These multiple streams of data are fused together and analysed respectively for ground truth via **SRGANs** & **DCNNs**.

HONOURS AND AWARDS

[09/2024] BobbleAI Keyboard, Gurgaon, Haryana

BobbleAI Datathon Winner Processed raw product data, i.e. [BobbleAI Keyboard](#)(70M+ users) event records in the last 5 days before July 1, 2024,i.e., about 8 million rows. Extracted & transformed JSON, to preprocess, ML-filter and store as structured **Apache Parquet** on **AWS S3** via serverless **AWS Lambda** functions & **PySpark** on **AWS Glue**.

Link: <https://github.com/DivitMittal/Datathon-BigData>

[10/2023] Alliance University, Anekal, Karnataka

Proglint's Computer Vision 2k23 National Hackathon Runner-ups

- Leverage the high-fidelity **CARLA** simulator to create diverse driving scenarios for an **autonomous vehicle driving system** with RGBA & LiDAR sensors. Utilizing deep learning techniques, i.e. **CNNs** for **semantic segmentation** to improve the vehicle's understanding of its surroundings, & manoeuvre.
- A **Driver Drowsiness Detection** system that uses **Siamese Neural Networks** and computer vision techniques, like **haar cascade** wave filters to monitor a driver's facial features in real-time and detect signs of fatigue. By analyzing eye states, & mouth movements, the system identifies potential drowsiness and enhances road safety.

Links: <https://github.com/DivitMittal/CARLA-Autonomous-Driving> | <https://github.com/DivitMittal/Driver-Drowsiness-Detection>

[01/2020] The Science Olympiad Foundation, New Delhi

International Rank 7 at the International Mathematics Olympiad

[12/2019] The British Council Division, British High Commission, New Delhi

International Rank 49 at the International English Olympiad (IEO)